Mass production of VCSELs for datacom and sensing applications

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Abstract: This paper reports the demonstration of our ability to realize precise process control and advanced process development. These abilities are pivotal for the VCSEL foundry to provide a reliable process and help achieving mass production. © The Author(s)

1. Introduction

Since the first demonstration of the vertical-cavity surface-emitting lasers (VCSELs) in 1978 [1], this pioneering invention has attracted much attention in many fields. VCSELs with low power consumption, high power-conversion efficiency, and small diversion angle shows the great potential in many applications, including LiDARs, time-of-flight (ToF), and proximity sensors [2]. Recently, VCSELs also exhibit an extremely large bandwidth, which is in excess of 35GHz [3]. The high-speed characteristic can support VCSELs to operate over 100Gb/s with 4-level pulse amplitude modulation (PAM-4) and even push to 200Gb/s per lane. This breakthrough makes VCSELs fulfill the demands of artificial intelligence (AI) and high performance computing (HPC) in datacenters. However, to maintain the outstanding performance of VCSELs and realize mass production, a precise and stable process control is pivotal for VCSEL fabrication. WIN Semiconductors is the leading foundry of III-V compound semiconductors and is famous for providing stable, precise, and advanced wafer-level process services.WIN has sufficient experience to fabricate VCSELs. Several VCSEL products have achieved mass production at WIN. When it comes to the VCSEL fabrication, a wet-oxidation process is always utilized to define the aperture of VCSEL after trench etching. The size of oxidation aperture (OA) is directly related to the optical and electrical characteristics of VCSELs. Consequently, the key to maintaining the performance of VCSELs stably is to precisely control the oxidation process and etching depth of trench. Furthermore, the thickness of silicon nitride (Si_xN_y) deposited by PECVD on the emitter region is also crucial for the high-speed performance of datacom VCSELs, which is highly related to the photon lifetime and will affect the VCSEL bandwidth significantly [4].

In this paper, we present the ability of WIN to control the VCSEL process. Hundreds of oxidation, Si_xN_y thickness, and trench-etching data are collected and demonstrated to exhibit our ability to achieve precise process control, which is beneficial for helping more VCSEL products to realize mass production. Additionally, WIN keep optimizing our critical process. By aging in a specific stress condition, the fine-tuned oxidation recipe shows a robust reliability performance, indicating that WIN has excellent ability of process development.

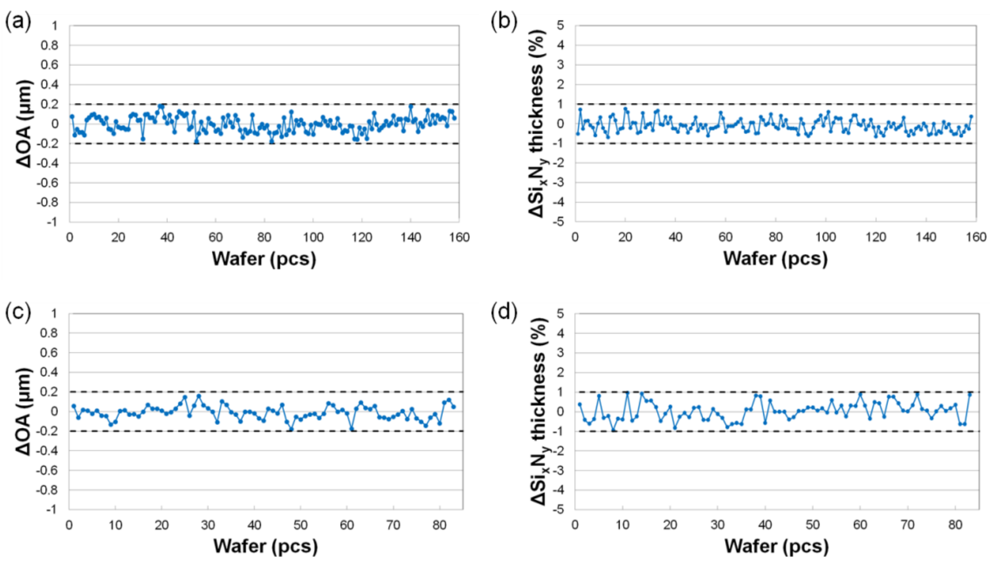

Fig. 1. Process monitoring charts of (a,b) a trench-type VCSEL and (c,d) a ring-type VCSEL. ΔOA and ΔSi_xN_y thickness represent the variation of OA diameter and the thickness of Si_xN_y, respectively.

2. Precise process control and advanced process development

As a III-V compound semiconductors foundry with decades experience of mass production in process and epitaxy, every step will be inspected and monitored systematically for stability and quality guarantee. For the epitaxial inspection in in-house epitaxial fab, x-ray, electrochemical capacitance voltage (ECV), and secondary ion mass spectrometer (SIMS) will be utilized to monitor not only lattice quality but also composition accuracy of the epitaxial structure. The particle level, defect counts, haze, Fabry–Pérot dip, etc., will also be inspected by surface scanning and photoreflectance measurement respectively. Additionally, to ensure the Al content of the critical oxidation layer, a wet-oxidation process would be performed in diffusion furnace by at least one quarter epitaxial wafer to obtain and confirm oxidation rate. At process line, for a consistent oxidation process, one pilot run EPI wafer from each epitaxial run will be input at first before processing the rest of EPI wafers, which helps oxidation process stability and repeatability. In addition to the oxidation, the optical and electrical characteristics of VCSELs are also highly related to the trench-etching depth and Si_xN_y deposition, there are strict inspections in these crucial process steps, such as the inspection of the trench-etching depth and Si_xN_y deposition thickness. Both critical processes are controlled by end-point mode and feed-forward recipe system, respectively. The systematic methodology and inspection can help us to provide a reliable process for every VCSEL applications. Figure 1 illustrates the statistics of the variation of the OA diameter and Si_xN_y thickness of two types of VCSELs in continuous process runs. An infrared microscope was used to monitor the OA diameter of VCSELs. Figure 1(a) and (b) are the process monitoring charts of trench-type VCSELs, whose target OA diameters are larger than 10μm. Figure 1(c) and (d) belong to ring-type VCSELs, whose target OA diameters are smaller than 10μm. Their target Si_xN_y thicknesses all exceed 5000Å. According to these statistics, no matter which type of VCSELs is, both OA diameter and Si_xN_y thickness can be well-controlled in our process. The variation of the OA diameter is in a range of ±0.2μm, and the Si_xN_y thickness fluctuates within lower than 2%. Figure 2(a) and (b) illustrate the statistics of the variation of the trench-etching depth of two types of VCSELs in continuous process runs. The variation of the trench-etching depth can be also well-controlled in a ±0.2μm range, exhibiting the ability of stable process control. Figure 2(c) shows our trench etching profile using the newly developed etching recipe. Notably, based on our advanced trench process development, we aim to control the target etching depth, and the trench footing is smaller than 1 pair of DBR. The precise etching technique can support datacom VCSELs to bury more AlAs in nDBR, which could be beneficial for improving the thermal characteristic of VCSELs [5].

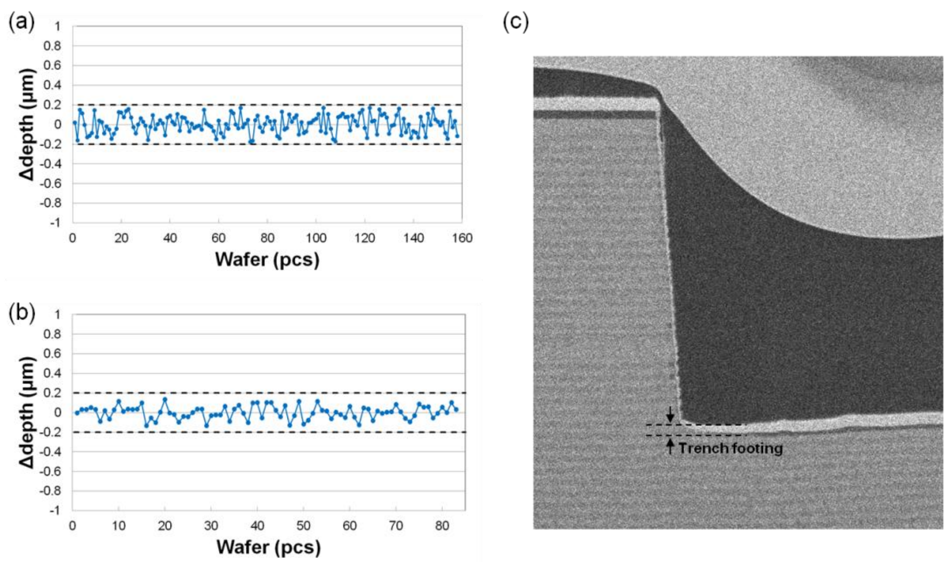

Fig. 2. Process monitoring charts of (a) a trench-type VCSEL and (b) a ring-type VCSEL, where Δdepth represents the variation of trench-etching depth. (c) The FIB image of the trench-etching profile using the newly developed etching recipe.

Additionally, because of the increasing demands of VCSELs in many novel applications, advanced process development is also an important ability for a VCSEL foundry supplier. Except for the trench-etching development, WIN keep optimizing our oxidation recipe, aiming to improve the robustness of VCSELs for novel applications. Figure 3 illustrates the reliability test of VCSELs with two different developed oxidation recipes. The OA diameters of these VCSELs are controlled at 10μm, and VCSEL chips are all packaged in TO headers. A strict stress condition is applied at 150^{o}C and 16.5mA. These two oxidation recipes are constituted by different oxidation conditions with different pretreatment conditions. In the first 100 hours, obvious power degradation can be observed in one VCSEL chip with oxidation recipe 1. When arriving 480 hours, all VCSEL chips with oxidation recipe 1 have power degradation between 10% and 45%. On the contrary, VCSELs with oxidation recipe 2 shows a robust reliability performance, as shown in Fig. 3(c) and (d). By this comparison, a newly developed oxidation recipe exhibits a clear development direction of the oxidation process. WIN will keep optimizing our oxidation process for providing a more reliable VCSEL foundry.

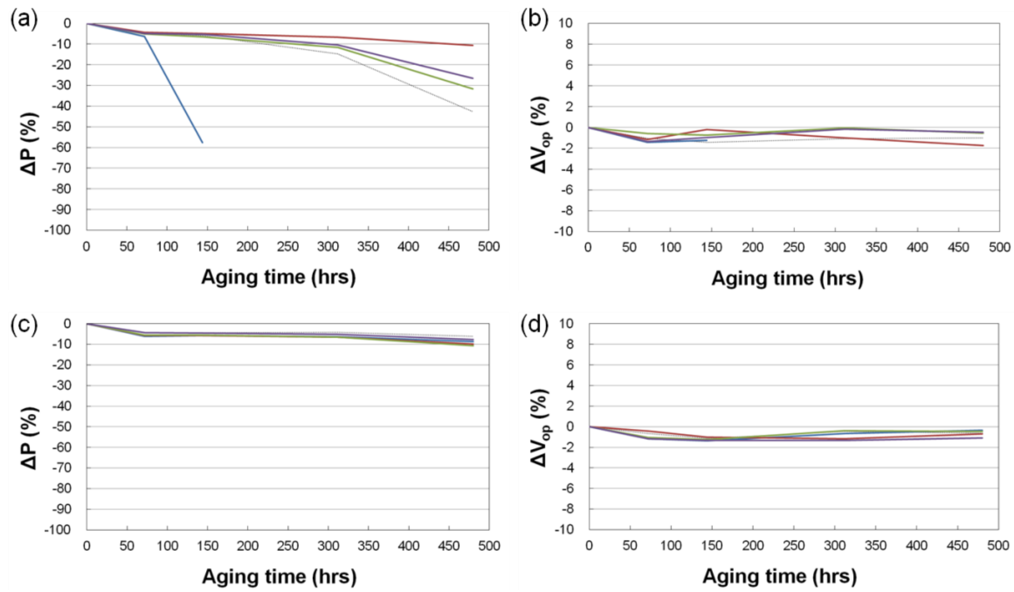

Fig. 3. Reliability test in different oxidation recipes. The power degradation (ΔP) and voltage increment (ΔV_{op}) of VCSELs with (a,b) oxidation recipe 1 and (c,d) oxidation recipe 2.

3. Conclusion

In this paper, we have demonstrated the ability of precise process control and advanced process development, which are crucial for mass production of VCSELs. By collecting hundreds of data, the OA diameter, Si_xN_y thickness, and trench-etching depth can be all well-controlled in a small range, exhibiting that WIN can provide a stable and reliable process for the VCSEL foundry. These characteristics can help more datacom or sensing VCSEL products to achieve mass production. Additionally, by aging in a strict stress condition, VCSELs with a new oxidation recipe show robust reliability performance, making our VCSEL foundry more reliable.

4. References

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